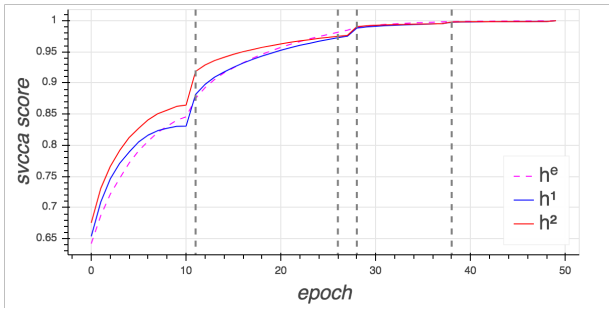

Figure 3: SVCCA score between representations at each epoch and from the final trained LM.

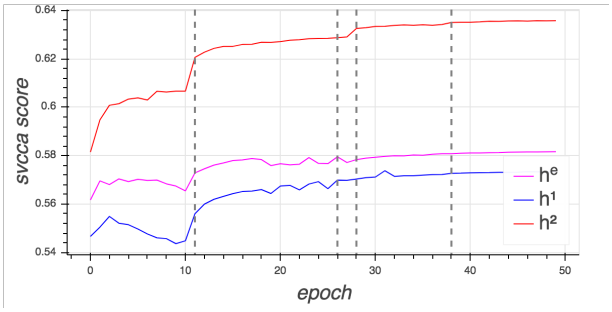

Figure 4: SVCCA score between the LM at each epoch and a LM with different initialization.

UDP silver POS and topic information use the same corpus, taken from the 100 longest articles in Wikipedia randomly partitioned in a 70/10/20 train/dev/test split. Each token is tagged with POS and with the ID of the source article. The corpus is taken from the LM training data, which may increase the similarity between the tag model and LM. Because both tag predictors are trained and tested on the same domain as the LM, they can be easily compared in terms of their similarity to the LM representation. Though the SEM corpus and the PTB corpus are different domains from the Wikipedia training data, we compare their activations on the same 191K-token 100-article test corpus.

Table 2 describes the training and validation corpus statistics for each tagging task. Note that topic and UDP POS both apply to the same en-wikipedia corpus, but PTB POS and SEM use two different unaligned sets from the GMB corpus.

4 Experiments, Results, and Analysis

A benefit of SVCCA is its flexibility: it can compute the correlation of a hidden representation to any other vector. Raghu et al. (2017) used it to understand learning dynamics by comparing a learned representation to snapshots of the same representation at different epochs during training. We use a similar experiment to establish the basic learning dynamics of our model. In our shallow 2-level model, activations at h^1 converge slightly after h^2 (Figure 3). This differs from the results of Raghu et al. (2017), who found that a 5-layer stacked LSTM LM exhibits faster convergence at lower layers, but this difference may be attributed to our much larger training data, which gives our model sufficient training data at early epochs.

Empirical upper bounds. Our main experiments will test the rate at which different linguistic categories are learned by different layers, but to interpret the results, we need to understand the behaviour of SVCCA for these models. In theory, SVCCA scores can vary from 0 for no correlation to 1 for perfect correlation. But in practice, these extreme cases will not occur. To establish an *empirical* upper bound on correlation, we compared the similarity at each epoch of training to the frozen final state of a LM with identical architecture but different initialization, trained on the same data (Figure 4).[2] The correlations increase over time as expected, but to a maximum near 0.64; we don't expect correlations between our LM and other models to exceed this value. We explore corresponding lower bounds in our main experiments below.

Correlations between different layers. Next we examine the correlation between different layers of the same model over time (Figure 5). We observe that, while over time correlation increases, in general closer layers are more similar, and they are less correlated than they are with the same layer of a differently initialized model. This supports the idea that we should compare recurrent layers with recurrent layers because their representations play similar roles within their respective architectures.

SVCCA vs. Diagnostic classifiers A popular method to analyze learned representations is to use a *diagnostic classifier* (Belinkov et al., 2017), a separate model that is trained to predict a linguistic category of interest, y_t, from an arbitrary hidden layer h_t. Diagnostic classifiers are widely used (Belinkov et al., 2018; Giulianelli et al., 2018). But if a diagnostic classifier is trained on enough

[2]This experiment is similar to the comparisons of randomly initialized models by Morcos et al. (2018).

tag	corpus	number of classes	token count train	token count dev	label $t+1$ acc	label $t+1$ ppl	label t acc	label t ppl	randomized acc	randomized ppl
UDP POS	wiki	17	665K	97K	50	4.3	93	1.2	21	8.9
PTB POS	GMB	36	943K	136K	51	4.7	95	1.18	14	18.0
SEM (coarse)	GMB	14	937K	132K	55	3.5	91	1.3	22	9.0
SEM (fine)	GMB	67	937K	132K	50	5.6	88	1.45	17	21.5
topic	wiki	100	665K	97K	36	19.1	37	16.3	5	81.6

Table 2: Tag predictor and tagger statistics. Accuracy and perplexity on $t+1$ are from the target tag predictor, on t are from the input tagger. Metrics obtained when training on randomly shuffled labels are provided as a low baseline. Accuracy is on the test set from the training domain (GMB or Wikipedia).

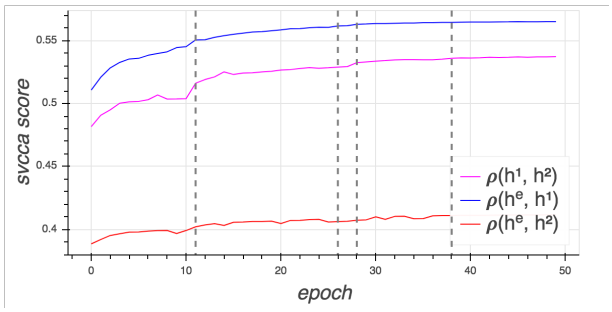

Figure 5: SVCCA score between different layers of the LM at each epoch. For example, $\rho(h^2, h^1)$ compares the activations h^2 with the activations h^1.

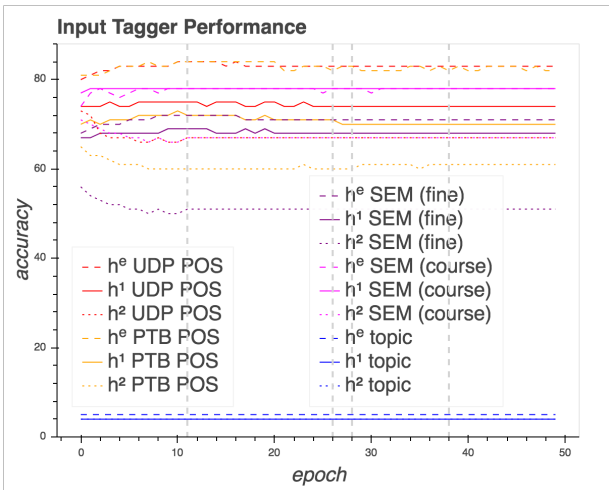

Figure 6: Learning dynamics interpreted with diagnostic classifiers labeling input word tag y_t.

examples, then random embeddings as input representations often outperform any pretrained intermediate representation (**?**Zhang and Bowman, 2018). This suggests that diagnostic classifiers may work simply by memorizing the association between an embedding and the most frequent output category associated with that embedding; since for many words their category is (empirically) unambiguous, this may give an inflated view of just how much a model "understands" about that category.

Our use of SVCCA below will differ from the use of diagnostic classifiers in an important way. Diagnostic classifiers use the intermediate representations of the LM as inputs to a tagger. A representation is claimed to encode, for example, POS if the classifier accurately predicts it—in other words, whether it can *decode* it from the representation. We will instead evaluate the *similarity* between the representations in an LM and in an independently-trained tagger. The intuition behind this is that, if the representation of our LM encodes a particular category, then it must be similar to the representation of model that is specifically trained to predict that category. A benefit of the approach is that similarity can be evaluated on *any* dataset, not only one that has been labeled with the linguistic categories of interest.

Another distinction from the typical use of diagnostic classifiers is that probes are usually used to decode tag information about the context or most recent *input* from the hidden state at the current step. Because the hidden representation at time t is meant to encode predictive information about the target word at time $t+1$, we treat it as encoding a prediction about the tag of the *target* word.

To understand the empirical strengths and weaknesses of these approaches, we compare the use of SVCCA and diagnostic classifiers in understanding learning dynamics. In other words, we ask: is our first conceptual shift (to SVCCA) necessary? To test this, we use the same model as Belinkov et al. (2017), which classifies an arbitrary representation using a ReLU followed by a softmax layer. To be consistent with Belinkov et al. (2017), we use y_t as their target label. We repeat their method in this manner (Figure 6) as well as applying our second modification, in which we instead target the label y_{t+1} (Figure 7).

We found the correlations to be relatively stable over the course of training. This is at odds with the results in Figures 2 and 3, which suggest that

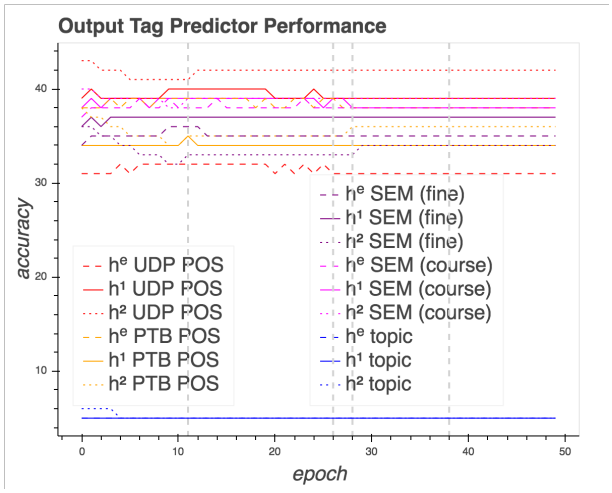

Figure 7: Learning dynamics interpreted with diagnostic classifiers labeling target word tag y_{t+1}.

representations change substantially during training in ways that materially affect the accuracy of the LM. This suggests that diagnostic classifiers are not illustrating improvements in word representations throughout training, and we conclude that they are ineffective for understanding learning dynamics. Our remaining experiments use only SVCCA.

4.1 SVCCA on Output Tag Prediction

We applied SVCCA to each layer of our LM with the corresponding layer of each tag predictor in order to find the correlation between the LM representation and the tag model representation at each level (Figure 8). To establish empirical lower bounds on correlation, we also trained our taggers on the same data with randomly shuffled labels, as in Zhang et al. (2016). These latter experiments, denoted by the dotted lines of Figure 8, show how much of the similarity between models is caused by their ability to memorize arbitrary associations. Note that the resulting scores are nonzero, likely because the linguistic structure of the input shapes representations even when the output is random, due to the memorization phase of training (Shwartz-Ziv and Tishby, 2017).

The strongest similarity at recurrent layers belongs to the most local property, the UDP POS tag. Both coarse- and fine-grained semantic tags, which rely on longer range dependencies, fall below UDP POS consistently. Topic, which is global to an entire document, is the least captured and the slowest to stabilize. Indeed, correlation with true topic falls consistently *below* the score for a model

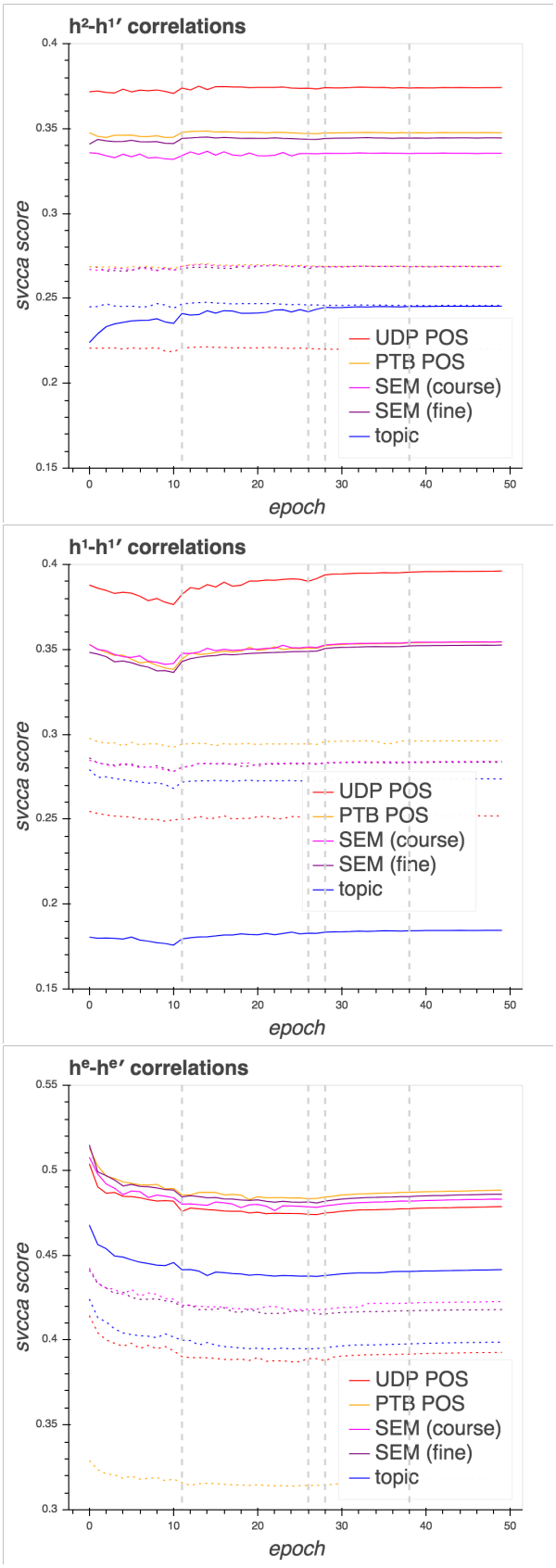

Figure 8: SVCCA correlation scores between the LM predicting x_{t+1} and the tag model predicting y_{t+1}. At the end of each epoch, we compare the current LM with the final tag model. Dotted lines use shuffled tags. Gray vertical lines mark when the step size is rescaled.